

BRYAN TANG

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Education

The University of Texas at Austin

August 2023 – May 2027

Bachelor of Science in Computer Science and Mathematics

Austin, Texas

Minor in Business

GPA: 3.6

Experience

Capital One

June 2026 – August 2026

Software Engineer Intern

Richmond, Virginia

- Engineered a **Go AWS Lambda** that validates, tokenizes, and submits payment files, using **S3** as a durable state machine with downstream integrations behind an **OAuth2** gateway.
- Cut **S3** object-deletion latency by 50% by parallelizing independent delete operations with bounded **goroutines**.
- Developed a **Node.js** UI surfacing real-time pipeline status (success, failure, and in-progress state for each run) built and tested against **LocalStack** to give operations teams end-to-end visibility into file processing.

Hewlett Packard Enterprise

May 2025 – May 2026

Software Engineer Intern - Part-time co-op

Spring, TX

- Wrote a **Cryptography Bill of Materials (CBOM)** generator that scans **Docker** containers with **Syft** and fuzzy string matching to inventory crypto assets and flag quantum-vulnerable algorithms for post-quantum migration.
- Architected an open-source **Go CLI** that queries server hardware over the **Redfish** API and runs automated firmware auditing and validation pipelines, replacing manual inspection workflows and cutting audit time by 40%.
- Migrated the tool's backing store from flat JSON to an indexed **SQLite** database, reducing firmware-record lookup latency 65% and scaling it to enterprise inventory sizes.

Applied Research Laboratories

January 2025 – May 2025

Software Engineer Intern

Austin, TX

- Built a **Java Swing** application with asynchronous, fault-tolerant data fetching and JSON parsing to visualize real-time sensor data via **REST API**, reducing data retrieval latency by 50%.

PCOnline

May 2024 – August 2024

Software Engineer Intern

Newark, DE

- Designed a full-stack **Python** application with a **Tkinter** GUI to automate label generation for 6000+ returned products, scaling warehouse reconciliation processes.

Projects

POSIX Shell Implementation | C, C++

- Created a **POSIX**-compliant shell that supports 30+ commands, scripting, redirection, pipelining, and variables.
- Implemented a custom tokenizer and abstract syntax tree to parse logical operators and enable short-circuiting.

AI Firmware Vulnerability Triage Platform | Python, Go, FastAPI, LLM, scikit-learn

- Designed a modular three-stage inference pipeline: a **RAG** retrieval layer (sentence-transformer embeddings over a document corpus), a **gradient-boosted ML** scoring model, and a tool-calling **LLM** agent.
- Served results through a **FastAPI** backend and **Go CLI**; applied the system to auto-triage and prioritize security findings into P0–P4 tiers.

Operating System | C, C++, x86

- Wrote a minimal **Unix-like** operating system kernel in **C/C++** and **x86** assembly.
- Implemented core OS components including process management, **ELF** binary loading, **system call** handling, memory management (**VMM**), synchronization primitives (semaphores, barriers), and basic file system support (**ext2**).

Technical Skills

Languages: Python, Java, C, C++, HTML/CSS, Go, SQL, JavaScript, Bash

Technologies: Linux, Git, Docker, Kubernetes, REST API, Redfish API, AWS, PostgreSQL, SQLite

Concepts: Operating Systems, Computer Architecture, Systems Engineering, Software Engineering, Automation/Scripting

Work Eligibility: US Citizen